

# E-Wave Plane-Wave Scattering by Sinusoidal Open PEC Strips via the Method of Discrete Singularities

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**Abstract**—We present a full-wave numerical study of plane-wave E-wave (TM) scattering by two-dimensional open sinusoidal perfectly-electric-conductor (PEC) strips. The weighted SIE is discretized by a Chebyshev-Nystrom form of the method of discrete singularities (MDS), after analytically extracting the inverse-square-root edge behavior of the current. A flat-strip benchmark provides a consistency check: boundary residuals are near  $10^{-13}$ , while the total scattering width reaches 40.0 wavelengths at  $N = 200$ . For sinusoidal corrugation the far-field develops finite-strip lobes near the infinite-periodic Floquet reference angles. A parametric study over amplitudes from 0.5 to 2.5 wavelengths and spatial frequencies from 0.10 to 0.30 inverse wavelengths shows that amplitude controls the relative strength of off-normal lobes while spatial frequency sets their angular positions. Across the 25 sinusoidal cases, the total scattering width spans 38.940–40.656 wavelengths (a 4.3% relative range about the mean), so corrugation changes both angular distribution and total scattering.

**Index Terms**—method of discrete singularities, sinusoidal open strip, plane-wave scattering, finite-strip scattering, E-wave, PEC strip

## I. INTRODUCTION

Corrugated metallic surfaces are ubiquitous in microwave engineering. The sinusoidal open strip is a particularly compact example: it serves as a meander-line delay element [1], slow-wave resonator, and unit cell of high-temperature superconducting (HTS) bandpass filters [2], [3]. At microwave frequencies, HTS films satisfy  $k\lambda_L \lesssim 10^{-4}$  (London penetration depth  $\lambda_L \sim 100$  nm), so the PEC boundary condition is an excellent approximation.

When a plane wave is incident on such a strip, the corrugated surface couples the specularly reflected field into additional off-normal lobes at angles set by the sinusoidal period. Here a *lobe* means a local maximum of the angular far-field amplitude. The corresponding infinite periodic continuation is described by Floquet theory [4], [5], but a finite strip with few periods has significant endpoint diffraction that demands a full-wave treatment.

Among rigorous 2-D methods for open-arc scattering, the classical *method of discrete singularities* (MDS) [6], [7] combines analytical edge-singularity extraction with Chebyshev

Nystrom discretization. The implementation used here follows the differentiated MDS system: the current endpoint singularity is removed analytically, the first-kind SIE is differentiated into a Cauchy-type equation, and one supplementary integral condition restores the lost constant. The resulting  $N \times N$  dense system is solved by Gaussian elimination; the reported residuals measure algebraic satisfaction of this discrete system. MDS has been validated for parabolic multireflector antennas with complex-source-point (CSP) beams [6], [8]; related open-strip and reflector formulations have also been treated with analytical regularization, edge-conditioned Nystrom schemes, and rapidly convergent SIE solvers [9]–[12]. Here we apply the same open-arc framework to plane-wave excitation and the finite sinusoidal strip geometry.

## II. PROBLEM FORMULATION

### A. Boundary-value problem and SIE

We seek the scattered E-wave field  $U$  when a 2-D PEC arc  $\Gamma$  is illuminated by a unit-amplitude plane wave. The common Euler time factor  $e^{i\omega t}$  is omitted, so the phasor is

$$U_0(\mathbf{r}) = e^{-ik(x \cos \beta + y \sin \beta)}, \quad (1)$$

with  $k = 2\pi/\lambda$ , all lengths in wavelengths ( $\lambda = 1$ ,  $k = 2\pi$ ), and  $\beta = \pi/2$  (normal incidence, wave propagating in the  $+y$  direction). The total field is  $U_{\text{tot}} = U_0 + U$ , where  $U_0$  is the incident field and  $U$  is the scattered field. The scattered field satisfies the Helmholtz equation outside  $\Gamma$ , the Sommerfeld radiation condition, local energy finiteness at arc endpoints, and the Dirichlet condition  $U|_{\Gamma} = -U_0|_{\Gamma}$ .

The single-layer representation of  $U$  leads, after parameterizing the arc as  $\mathbf{r}(t) = (x(t), y(t))$ ,  $t \in [-1, 1]$ , and factoring the inverse-square-root edge singularity into a smooth density  $v(t)$  via  $j|\mathbf{r}'(t)| = v(t)(1-t^2)^{-1/2}$ , to the first-kind SIE

$$\int_{-1}^1 H_0^{(1)}(kR(t, t_0)) v(t) \frac{dt}{\sqrt{1-t^2}} = -U_0(t_0), \quad (2)$$

where  $R(t, t_0) = |\mathbf{r}(t) - \mathbf{r}(t_0)|$ .

### B. MDS discretization, geometry, and Floquet references

After differentiation, the MDS system is discretized by weighted Chebyshev quadrature plus one supplementary row. With source nodes  $t_i = \cos[(2i-1)\pi/(2n)]$ ,  $i = 1, \dots, n$ , and second-kind collocation nodes  $\tau_j = \cos[j\pi/n]$ ,  $j = 1, \dots, n-1$ , the boundary rows are

$$\frac{1}{n} \sum_{i=1}^n \left[ \frac{1}{t_i - \tau_j} + K(t_i, \tau_j) \right] v(t_i) = f(\tau_j), \quad (3)$$

where  $K$  is the regularized derivative kernel and  $f = -dU_0/d\tau$ . For one strip the regularized kernel is evaluated as

$$K(t, \tau) = -kH_1^{(1)}(kR) \frac{\partial R}{\partial \tau} - \frac{1}{t - \tau}, \quad (4)$$

with

$$\frac{\partial R}{\partial \tau} = -\frac{[\mathbf{r}(t) - \mathbf{r}(\tau)] \cdot \mathbf{r}'(\tau)}{R(t, \tau)}. \quad (5)$$

Thus the singular part is handled analytically, while only a smooth remainder is sampled numerically. The supplementary row is

$$\frac{1}{n} \sum_{i=1}^n M(t_i) v(t_i) = c, \quad (6)$$

where  $M$  is the weighted single-layer integral of the kernel and  $c = -\int_{-1}^1 U_0(\tau) d\tau / \sqrt{1 - \tau^2}$ . For self interaction the logarithmic term in  $M$  is also extracted:

$$M(t_i) = \int_{-1}^1 \left[ H_0^{(1)}(kR) - \frac{2i}{\pi} \ln |t_i - \tau| \right] \frac{d\tau}{\sqrt{1 - \tau^2}} - 2i \ln 2. \quad (7)$$

Equations (3)–(6) are solved by Gaussian elimination. The scattered far-field pattern is evaluated from the asymptotic single-layer potential,

$$\Phi_{\text{sc}}(\varphi) \approx \frac{1}{n} \sqrt{\frac{2}{\pi i}} \sum_{i=1}^n e^{-ik[\cos \varphi x(t_i) + \sin \varphi y(t_i)]} v(t_i), \quad (8)$$

and the total 2-D scattering width is

$$\sigma = \frac{1}{\pi} \int_0^{2\pi} |\Phi_{\text{sc}}(\varphi)|^2 d\varphi. \quad (9)$$

This value follows the far-field normalization in (8) and is used here for relative comparison across the same numerical setup. The residual reported below is

$$\varepsilon_{\text{bc}} = \|\mathbf{A}\mathbf{v} - \mathbf{b}\|_{\infty}, \quad (10)$$

where  $\mathbf{A}\mathbf{v} = \mathbf{b}$  denotes the assembled discrete MDS system.

The sinusoidal strip with  $L = 10\lambda$  and phase zero is

$$\Gamma_s = \left\{ (x, y) : x = \frac{L}{2} t, \quad y = A \sin(2\pi \nu x), \quad t \in [-1, 1] \right\}. \quad (11)$$

For the infinite periodic continuation with period  $\Lambda = 1/\nu$ , normal-incidence Floquet theory gives propagating forward-hemisphere reference angles at

$$\varphi_m = \arccos(m\nu), \quad |m\nu| < 1. \quad (12)$$

The finite strip ( $L\nu$  periods) broadens these into finite-strip lobes whose centers are expected near (12). With  $N = 200$

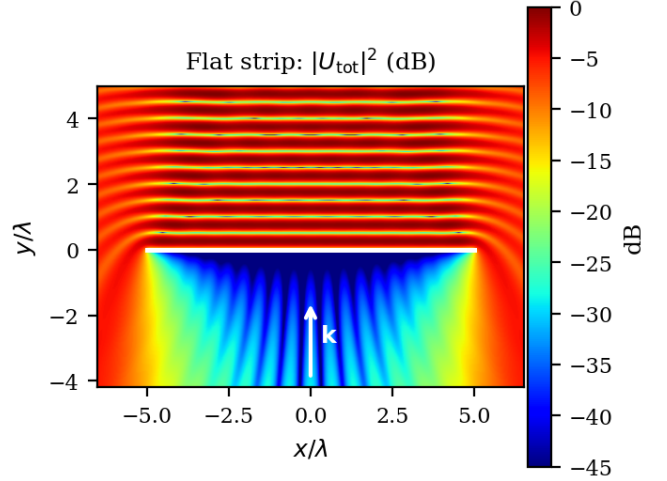


Fig. 1. Flat strip ( $A = 0$ ,  $L = 10\lambda$ ,  $N = 200$ ): normalized total near-field intensity  $|U_{\text{tot}}|^2$  in dB. The arrow marks the incident-wave direction.

TABLE I  
N-CONVERGENCE: TOTAL SCATTERING WIDTH  $\sigma$  FOR THE FLAT STRIP  
( $A = 0$ ,  $L = 10\lambda$ ,  $\beta = \pi/2$ ).

| $N$ | $\sigma$ ( $\lambda$ ) | Rel. change          | Residual              |
|-----|------------------------|----------------------|-----------------------|
| 32  | 39.857                 | —                    | $3.4 \times 10^{-14}$ |
| 64  | 39.999                 | $3.6 \times 10^{-3}$ | $2.7 \times 10^{-14}$ |
| 100 | 40.0                   | $2.8 \times 10^{-5}$ | $4.8 \times 10^{-14}$ |
| 150 | 40.0                   | $1.2 \times 10^{-6}$ | $1.1 \times 10^{-13}$ |
| 200 | 40.0                   | $2.2 \times 10^{-8}$ | $1.1 \times 10^{-13}$ |

nodes the worst-case arc length in the sweep ( $A = 2.5\lambda$ ,  $\nu = 0.30$ ) is  $\approx 32.31\lambda$ , giving about 6.2 nodes per wavelength along the strip. Near-field figures show the total field  $U_0 + U$  on a Cartesian grid, normalized to the maximum intensity in the plotted window. Far-field curves show  $20 \log_{10} |\Phi_{\text{sc}}|$  normalized to the peak of the corresponding pattern, so their purpose is angular comparison rather than absolute calibration.

## III. NUMERICAL RESULTS

### A. Flat strip benchmark

Setting  $A = 0$  gives a flat horizontal strip of length  $L = 10\lambda$ . The near-field and far-field benchmark plots are shown in Figs. 1 and 2; the boundary residual of the  $N = 200$  discrete system is  $1.10 \times 10^{-13}$ . The far-field pattern exhibits the expected two-lobe structure: the  $90^\circ$  lobe corresponds to forward diffraction (shadow region), and the  $270^\circ$  lobe to backward specular reflection, with weaker edge-diffraction contributions at  $0^\circ$  and  $180^\circ$ .

Table I shows  $N$ -convergence of  $\sigma$ . The sequence rapidly approaches  $40.0\lambda$ , while the residual remains near  $10^{-13}$  or better. This residual measures algebraic satisfaction of the discrete MDS system; convergence of the physical observable is assessed by the change in  $\sigma$ .

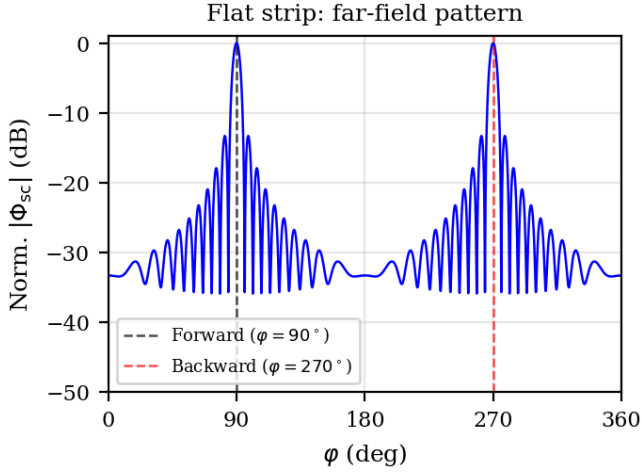


Fig. 2. Flat strip normalized scattered far-field amplitude. The dashed lines mark forward diffraction at  $90^\circ$  and backward specular reflection at  $270^\circ$ .

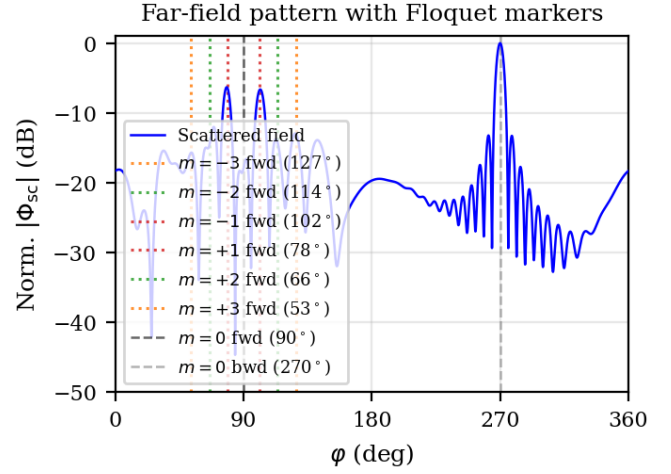


Fig. 4. Sinusoidal strip normalized scattered far-field amplitude. Vertical markers show infinite-periodic Floquet reference angles.

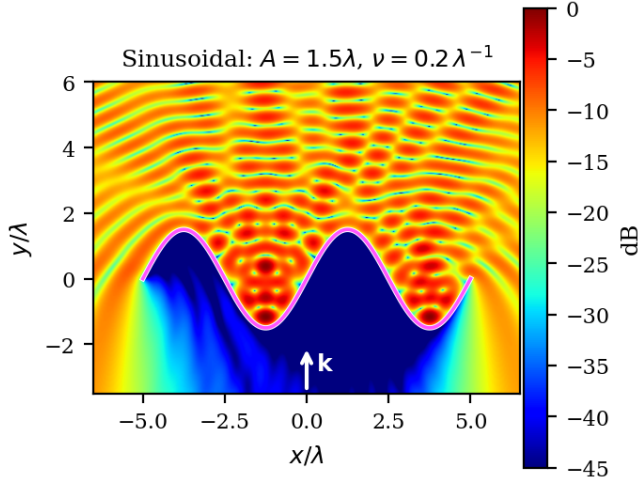


Fig. 3. Sinusoidal strip ( $A = 1.5\lambda$ ,  $\nu = 0.20 \lambda^{-1}$ ): normalized total near-field intensity with the strip profile overlaid.

Additional flat-strip checks at  $N = 400$  and  $N = 600$  changed the displayed value only beyond the shown precision. Thus  $N = 200$  is used as the working resolution for the sweep.

### B. Sinusoidal strip: finite-strip Floquet references

For  $A = 1.5\lambda$ ,  $\nu = 0.20 \lambda^{-1}$  ( $L\nu = 2$  periods) the boundary residual is  $1.62 \times 10^{-13}$  and  $\sigma = 40.011\lambda$ . The near-field in Fig. 3 shows that the corrugation disrupts the regular standing-wave pattern of the flat strip, producing complex interference structures above and below the arc. The corresponding far-field pattern is shown in Fig. 4.

The far-field in Fig. 4 reveals the key result: strong finite-strip off-normal lobes appear near the  $m = \pm 1$  reference angles  $78.5^\circ$  and  $101.5^\circ$ , with weaker  $m = \pm 2$  signatures near  $66.4^\circ$  and  $113.6^\circ$ . These angles are consistent with the Floquet references (12) for  $\nu = 0.20 \lambda^{-1}$ , while finite length broadens

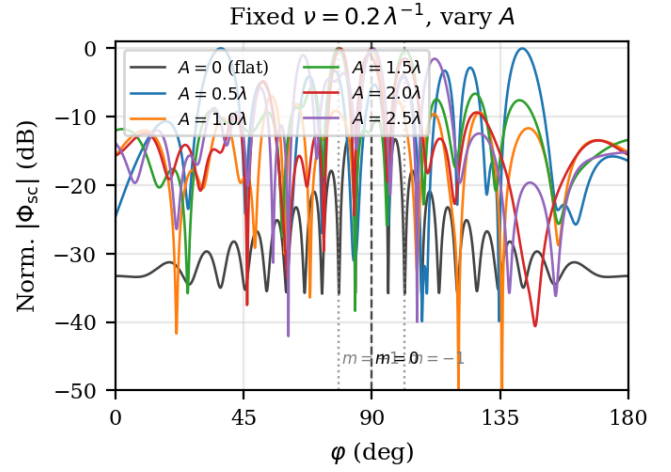


Fig. 5. Forward-hemisphere pattern evolution at fixed  $\nu = 0.20 \lambda^{-1}$  as  $A$  varies. Dotted verticals mark the  $m = \pm 1$  Floquet-reference angles.

and slightly shifts the maxima. The backward hemisphere shows a mirror structure. Although the finite strip ( $L\nu = 2$  periods) broadens each lobe, the marker positions remain useful references for identifying finite-strip lobe families.

### C. Parametric pattern evolution

Figures 5 and 6 summarize two parametric sweeps over the forward hemisphere ( $0^\circ$ – $180^\circ$ ).

**Amplitude dependence** [Fig. 5]: For the flat strip ( $A = 0$ ) a single dominant lobe sits at  $90^\circ$ . As  $A$  increases at fixed  $\nu = 0.20 \lambda^{-1}$ , strong lobes emerge near the  $m = \pm 1$  Floquet-reference angles ( $78.5^\circ$ ,  $101.5^\circ$ ), while the specular peak broadens and exchanges dominance with off-normal lobes. At  $A = 2.5\lambda$  the first-order lobes dominate the forward pattern, indicating substantial power transfer to off-normal directions.

**Frequency dependence** [Fig. 6]: With  $A = 1.5\lambda$  fixed, increasing  $\nu$  moves the  $m = +1$  reference angle from  $84.3^\circ$

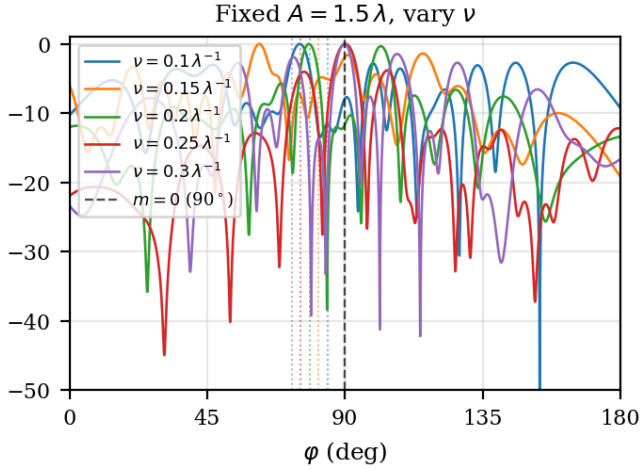


Fig. 6. Forward-hemisphere pattern evolution at fixed  $A = 1.5\lambda$  as  $\nu$  varies. Dotted verticals mark the  $m = +1$  reference angle for each spatial frequency.

( $\nu = 0.10$ ) to  $72.5^\circ$  ( $\nu = 0.30$ ), following (12). Because the finite strips contain only  $L\nu = 1\text{--}3$  periods, the observed maxima are broad and can be shifted from the infinite-periodic markers. At  $\nu = 0.25\text{--}0.30$  the  $m = \pm 2$  lobes enter the visible range. Across all 25 configurations the total scattering width  $\sigma$  spans  $38.940\text{--}40.656\lambda$ ; the range is  $1.715\lambda$ , or  $4.3\%$  of the mean value. The smallest value occurs at  $(A, \nu) = (2.0\lambda, 0.25\lambda^{-1})$ , and the largest at  $(2.5\lambda, 0.20\lambda^{-1})$ . All sweep residuals remain between  $7.1 \times 10^{-14}$  and  $4.1 \times 10^{-13}$ .

#### IV. DISCUSSION

The flat-strip result separates the finite-aperture baseline from the corrugation effects. For a horizontal PEC strip under normal incidence, the dominant forward and backward lobes are set by aperture diffraction and endpoint scattering. A sinusoidal profile mainly rephases the current density along the same open conducting arc. Thus the integrated width changes modestly in the sweep, while individual angular lobes can change by many decibels.

The Floquet markers should also be read carefully. They are exact angles for an infinite periodic continuation, whereas the simulated strip contains only  $L\nu = 1\text{--}3$  periods. The markers therefore organize broad finite-aperture maxima rather than define strict finite-strip diffraction orders. Endpoint contributions and local strip slope can shift the actual peak positions.

The current implementation is restricted to the PEC Dirichlet model. It therefore provides a clean reference case, while film thickness, kinetic inductance, dielectric loading, and losses should be treated later through impedance or transmission boundary conditions.

#### V. CONCLUSION

The MDS solver was applied to E-wave plane-wave scattering by sinusoidal open PEC strips. The solver was checked on a flat-strip benchmark: boundary residuals remained near  $10^{-13}$  and  $\sigma$  converged to  $40.0\lambda$  by  $N = 200$ . For the

sinusoidal strip, finite-strip far-field lobes were found near the infinite-periodic Floquet reference angles  $\varphi_m = \arccos(m\nu)$ , with finite-length broadening and small peak shifts. The parametric study identifies two independent design knobs: corrugation amplitude  $A$  controls the relative strength of off-normal lobes, and spatial frequency  $\nu$  sets the Floquet-reference angles. With the normalization used here, total scattering width is not invariant in the tested range: it spans  $38.940\text{--}40.656\lambda$  across the sinusoidal sweep. For superconducting meander-line devices these findings imply that the corrugation profile controls both the far-field scattering signature and the total scattered width, enabling shape-based crosstalk management in densely integrated microwave circuits. Planned extensions include H-wave polarization, oblique incidence, and impedance boundary conditions for resistive or finite-conductivity layers.

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